ELEC5305 Project Proposal

1 Title: Speech emotion recognition using acoustic feature extraction and machine learning

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3 Project Overview

This project aims to identify the emotion of the speaker by analysing their sound signal. This has many real-world applications including in teaching where it can be used to identify the engagements of students to particular teaching activities or materials and then improve them. It can also be used to improve AI audio interaction. If an AI is able to decipher the emotion of its user, it can provide information more specific to the speaker's situation instead of giving robotic responses.

The proposed solution uses acoustic feature extraction followed by supervised machine learning classification. Speech recordings will first be preprocessed and analysed to extract acoustic features that may vary between emotional states, such as signal energy, pitch and spectral characteristics. These features will be represented as a feature vector and provided to a machine learning classifier, which will predict the emotion category of the input speech signal. The performance of different acoustic feature representations will then be evaluated to investigate their effectiveness for speech emotion recognition.

4 Background and Motivation

Existing research in this topic area have shown that emotions can be expressed through speech and can be analyzed using acoustic and paralinguistic features. Examples of such features in this application are pitch, energy as well as spectral features including Mel-frequency cepstral coefficients (MFCCs). MFCC is a standard way of converting a short speech signal to a small set of numbers that describe its characteristic. When these features are fed into ML classifiers, an emotion value can be obtained from a defined set of possible emotions.

The primary challenge of these methods is choosing the appropriate features. Therefore, this project will test several acoustic feature sets and evaluate their classification performance to investigate which features are most effective for audio-based emotion recognition.

This topic was chosen because speech emotion recognition has potential applications in a range of real-world activities, making this topic quite relatable. The increasing use of artificial intelligence also makes the development of systems capable of analysing emotional characteristics in speech an interesting area of research. In addition, this project involves both acoustic signal processing and machine learning, providing the opportunity to learn and apply skills in both these areas.

5 Proposed Methodology

The signal processing tasks will be implemented primarily in MATLAB due to the availability of built-in audio and signal processing toolboxes, as well as relevant MATLAB signal processing code provided through Canvas. Python will be used for data preparation and machine learning classification using established packages such as Scikit-learn. Hugging Face will be investigated as a potential source of publicly available labelled speech datasets, while other sources such as Kaggle may also be considered. The selected dataset should contain a sufficient number of labelled speech recordings with a reasonably balanced distribution across the selected emotion categories.

After the speech recordings are loaded and preprocessed, they will be converted into a suitable format for analysis. Acoustic features will then be extracted from each speech signal. The initial feature sets will include basic acoustic features such as signal energy and pitch, spectral features, and Mel-frequency cepstral coefficients (MFCCs). Additional acoustic features identified in relevant research papers may also be investigated. The features extracted from each recording will be organised into feature vectors and normalised before being used for classification.

For the classification task, supervised machine learning models will be evaluated, preferably using Python and established packages such as scikit-learn, which provides access to a range of well-established classification algorithms. Different acoustic feature sets will be tested and their ability to classify the emotion expressed in previously unseen speech recordings will be evaluated. The performance of the different approaches will be compared using metrics such as classification accuracy and confusion matrices. The final system will take a speech recording as input, extract the selected acoustic features and predict the most likely emotion category.

6 Expected Outcomes

The expected outcome of this project is a working speech emotion recognition program that accepts a speech recording or a speech file as input (depending on practicality) and predicts the most likely emotion category based on extracted acoustic features. The program will be evaluated using previously unseen labelled speech recordings to determine its classification performance.

Performance metrics will include accuracy, precision, recall and F1-score. These results will be used to compare the effectiveness of the explored acoustic features, and will be used to choose which are the most appropriate for the final implementation.

The final deliverables will include the MATLAB and Python implementation, documented source code and a GitHub repository containing the project code. Also a report presenting the method and performance of the implementation will also be delivered.

7 Timeline (Weeks 1-13)

Week	Task
1-2	Consider Project Topic
3-5	Literature review and dataset collection
6-9	Initial implementation and testing.
10-11	Optimisation and evaluation
12-13	Final report and GitHub documentation

8 References

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9 Acknowledgement:

AI acknowledgement**:** ChatGPT (OpenAI, GPT-5.6 Luna) was used to assist with brainstorming the project topic, improving the structure and wording of the proposal, and explaining technical concepts. The final project decisions and content were reviewed and edited by the student.